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DSA0406

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(17)

CO1 - AT1 : Short Answers

1) What is Data Sci? Explain its primary objective?

⇒ It's a Multidisciplinary field with statistics, programming & etc.., its primary objective is to turn RAW Data into actionable knowledge.

2) Why is Data science is important in Today's digital world? State 4 reasons.

→ Enables data-driven decision-making systems, powers Automation, helps businesses understand customer, Drives various Innovations.

3) List any 4 benefits of Data science in business & society?

→ Beneficial recommendation systems, Improves decision making Accuracy, better customer targetting & personalisation, Early deduction of fraud / risks.

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94 %

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4) Mention 4 real-world applications of Data Science.

⇒ Real-world: ReCommandation systems, fraud detecting in banking, Medical diagnosis, traffic prediction.

5) What are the different facets of Data?

⇒ ◦ Structured : Organized in row/ Columns
◦ Unstructured : no fixed format
◦ semi-structured : partial organization
◦ Streaming Data : Continous real-time Data.

6) Differentiate between structured, semi-structured & unstructured data

⇒ Structures are fixed Schema like SQL DB, whereas semi-structured are as partial as XML files & unstructured data is non-organized like multimedia.

7) What is Big-Data? State its key characteristics (5Vs)

→ Big Data: Extremely large, Complex datasets beyond traditional processing capability.

- Five V's: Volume - Amount; velocity - speed; variety - types; veracity - Quality; value - usefulness.

8) Explain Big Data Eco-system.

→ Collection of tools / technologies to store, process & Analyse big data.

Examples: Hadoop (storage / processing), Spark, Kafka (streaming), MONGODB, Hive.

9) What are the Major Phases of the Data Science process?

→ Phases are: Data Retrieval → Data preparation (cleansing, integration) → Data Exploration / Analysis → Model building → Model evaluation → presentation / ~~dep~~ deployment of results.

10) What is Data Retrieval? Mention 2 Common Sources from which data can be retrieved?

⇒ The process of accessing / Extracting data from storage sources for analysis.

Sources: DB (SQL, No SQL), APIs, ~~IoT~~ web Scraping, flat files (CSV / Excel), Sensors.

11) Why is Data Collection is Considered an important step in DS process?

⇒ Forms the foundation for all analyses; poor-quality collected data leads to unreliable models; ensures relevant, sufficient data for Accurate insights.

12) What is Data Cleansing? List any 4 Common techniques?

⇒ Process of detecting / correcting errors, inconsistencies & missing values in data

Techniques

→ Handling Missing values
→ Removing Duplicates
→ Correction of inconsistent
→ Outlier detection

13) What is Data integration? Why is it necessary?

⇒ Combining data from multiple heterogeneous sources, necessary since integration enables comprehensive analysis & avoids siloed insights.

14) Explain Data Transformation with suitable Examples.

⇒ Conversion of data into a suitable format (or) structure for analysis.

Examples

→ Normalizations (scaling values)
→ encoding categorical variables
→ aggregation
→ Log transformation.

15) What is Data Analysis? Mention any 4 techniques?

⇒ Process of inspecting / modelling data to discover useful information & support conclusions.

Techniques:

→ Statistical Analysis → Clustering
→ Regression Analysis → Classification